CO2xN: Porewater Sulfide

Month Year Samples

2026-05-27

Code Set up

run information

```
##Before or After Reruns (YES or NO)
After_Reruns = "YES"
#If NO will not write final data file
  #And Will Write a Rerun File

###things that need to be changed

sample_month = "May"
sample_year = "2026"
Run_Date = as.Date("05/19/2026", format = "%m/%d/%Y")
Run_by = "Melanie Giessner"#Instrument user
Script_run_by ="Melanie"#Code user

Run_notes=""#any notes from run

#Import Plate Data

plates<- c("Plate1","Plate2","Plate3","Plate4","Plate5")

folder_path_rawdata <- paste0("Raw Data")
Raw_plates <- list.files(path = folder_path_rawdata, full.names = TRUE,pattern = "Plate")
head(Raw_plates)
```

```
## [1] "Raw Data/20260518_LTREB_H2S_Plate1.csv"
## [2] "Raw Data/20260518_LTREB_H2S_Plate2.csv"
## [3] "Raw Data/20260518_LTREB_H2S_Plate3.csv"
## [4] "Raw Data/20260518_LTREB_H2S_Plate4.csv"
## [5] "Raw Data/20260519_LTREB_H2S_Plate5.csv"
```

```
#Import Run Date for Each Plate for Std curve qaqc
Run_dates <- tibble(
    Plate = plates,
    Run_Date = c("05/18/2026","05/18/2026","05/18/2026","05/18/2026","05/19/2026")
  )

  Run_dates <- Run_dates %>% mutate(Run_Date = as.Date(Run_Date, format = "%m/%d/%Y"))


# Define the file path for QAQC log file - NO Need to change just check year
log_path <- "Processed Data/LTREB_2M_H2S_QAQC_2026.csv"
```

```
# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/GCReW_CO2xN_Porewater_H2S_202605.csv"
rerun_path <- "Processed Data/GCReW_CO2xN_Porewater_H2S_202605_reruns.csv"
```

##Read in metadata and create similar sample IDs for matching to samples

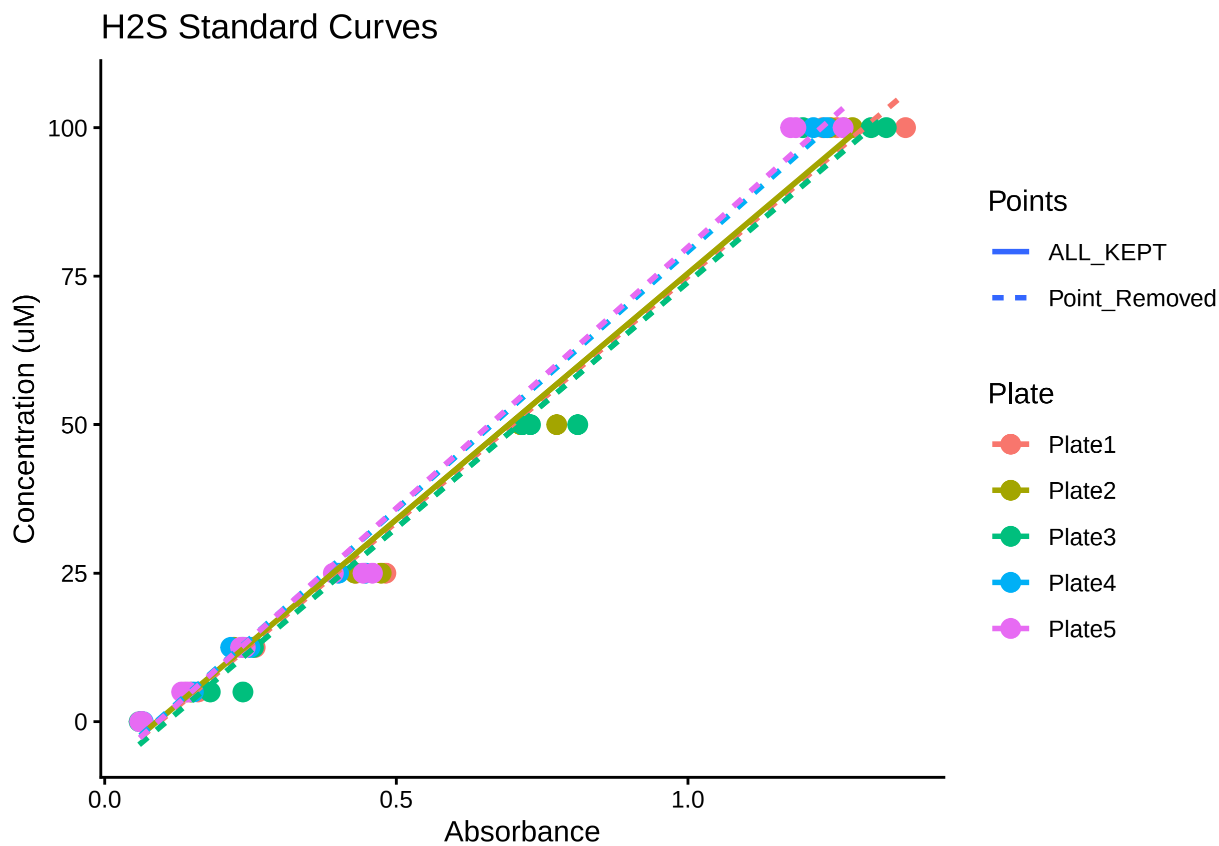
##Read in data

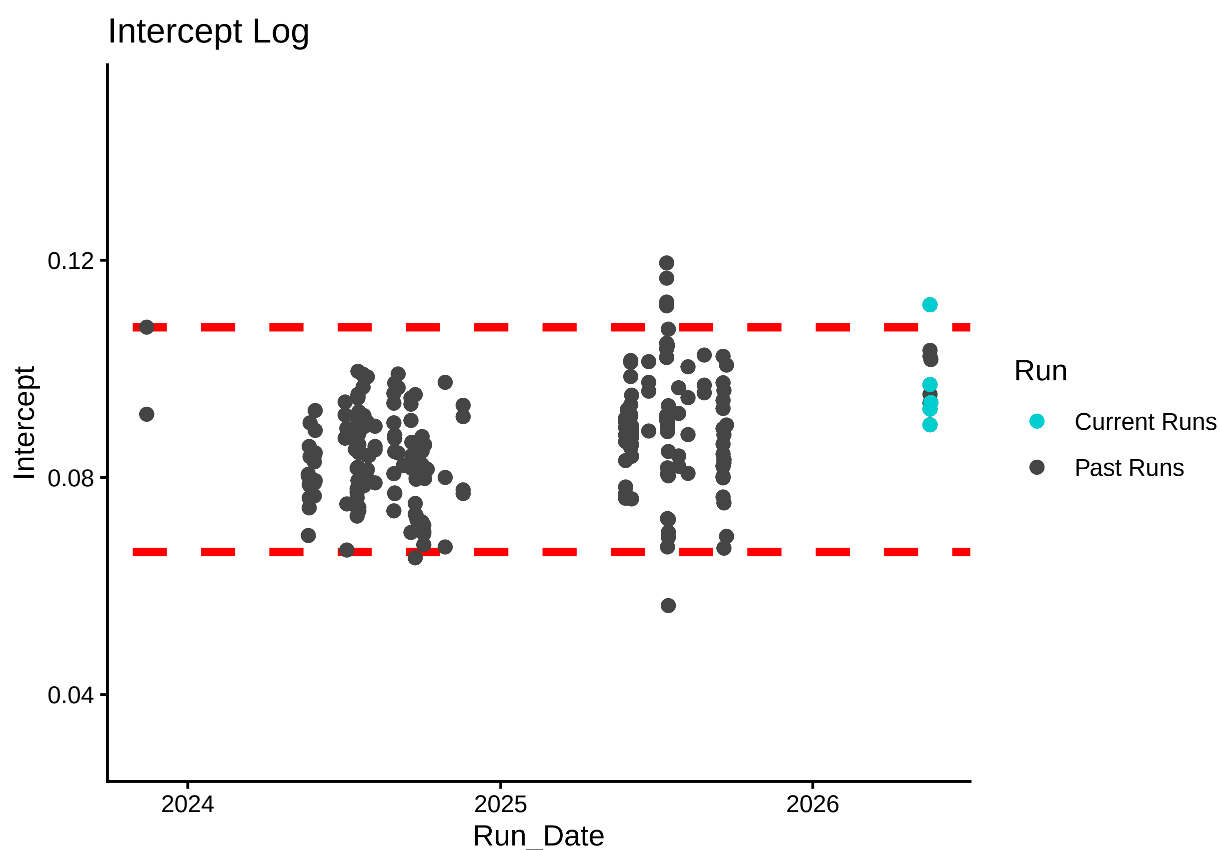
##Remove High CV Stds

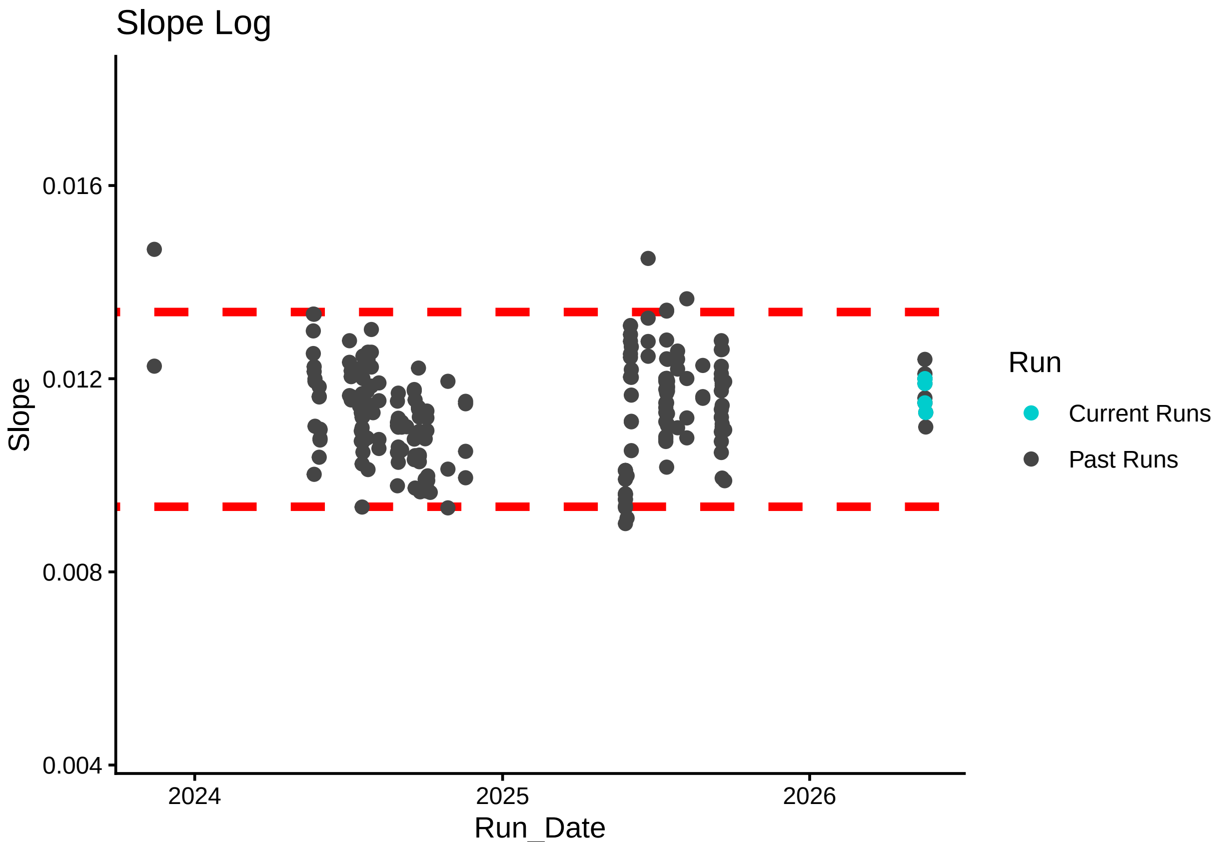
##Check R2 & Remove Standards if Necessary

```
## Joining with `by = join_by(Plate, IDs)`
```

Plot Standard Curves





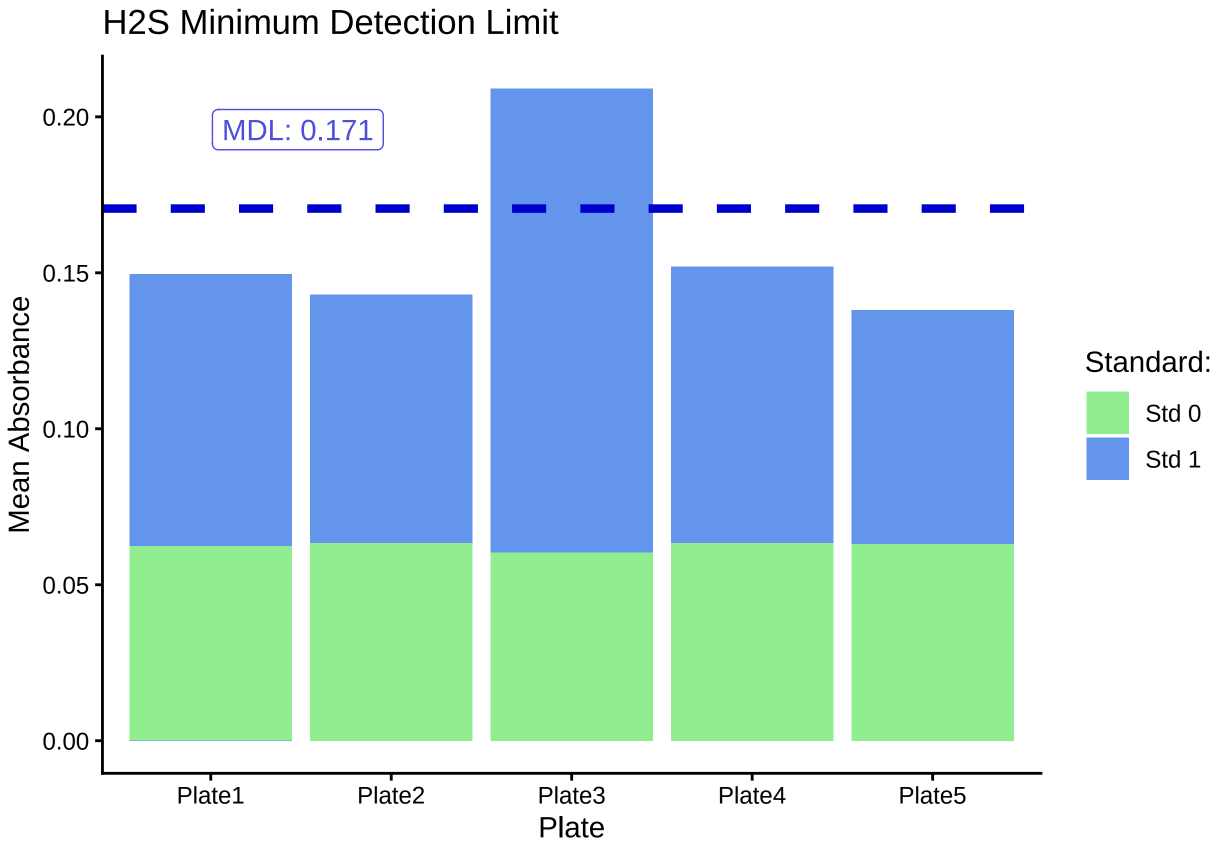


Return Flags For Standard Curve

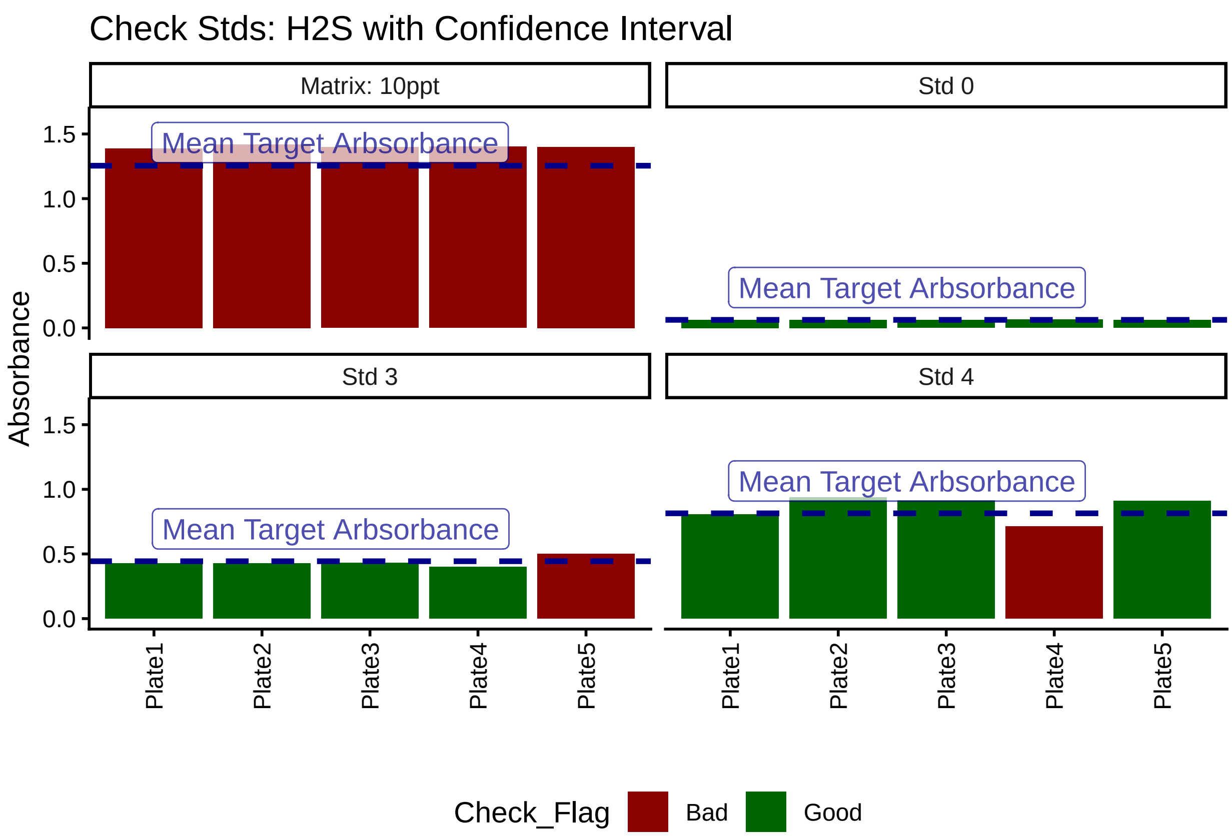
Plate	R2_Flag	CV_Flag
Plate1	R2_Good	CV_Good
Plate2	R2_Good	CV_Good
Plate3	R2_Good	CV_Good
Plate3	R2_Good	CV_High
Plate4	R2_Good	CV_Good
Plate5	R2_Good	CV_Good

Plate	Curve_RMVD
Plate1	KEPT
Plate2	KEPT
Plate3	REMOVED
Plate3	REMOVED
Plate4	KEPT
Plate5	KEPT

Method Detection Limit



Compare Check Standards to Standards



Display Any Check Flags

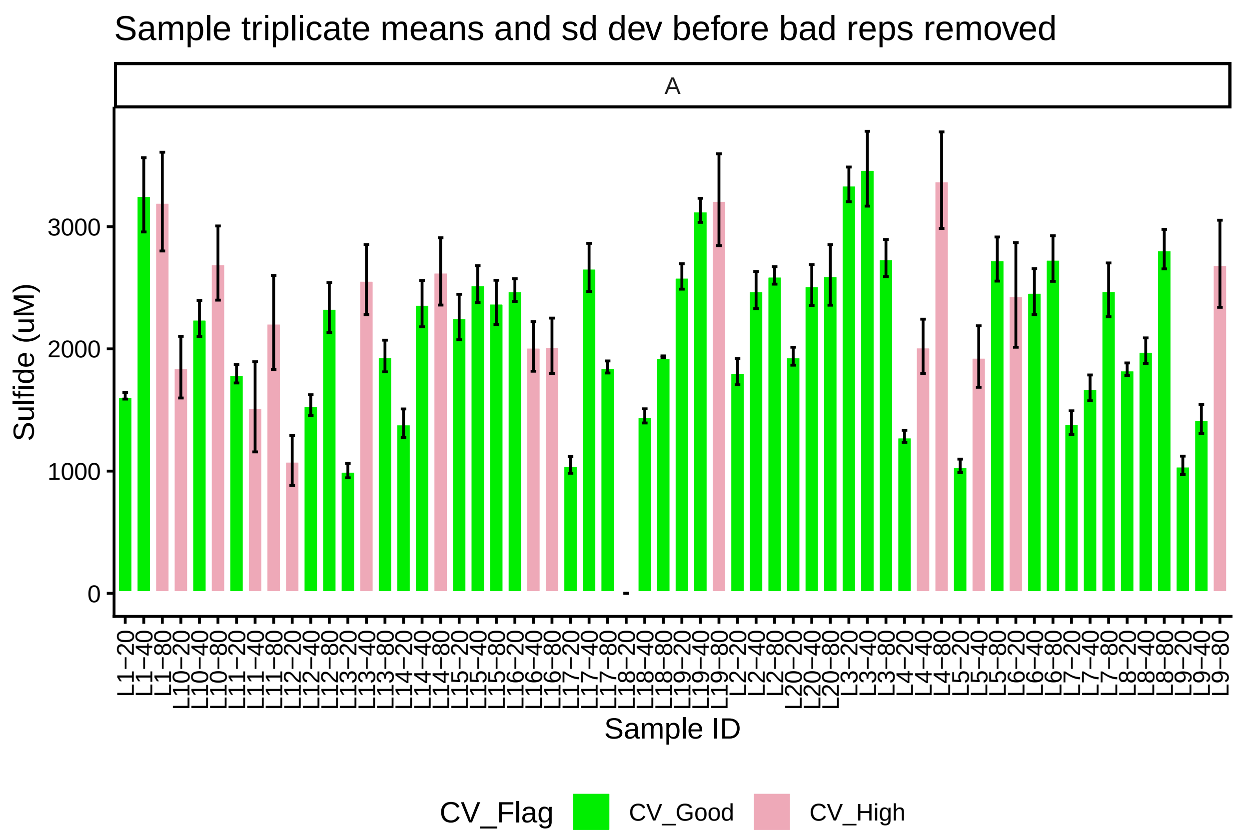
Plate	CHKs_Flag
Plate1	Std5Chk_Bad
Plate2	Std5Chk_Bad
Plate3	Std5Chk_Bad
Plate4	Std5Chk_Bad
Plate4	Std4Chk_Bad
Plate5	Std3Chk_Bad
Plate5	Std5Chk_Bad

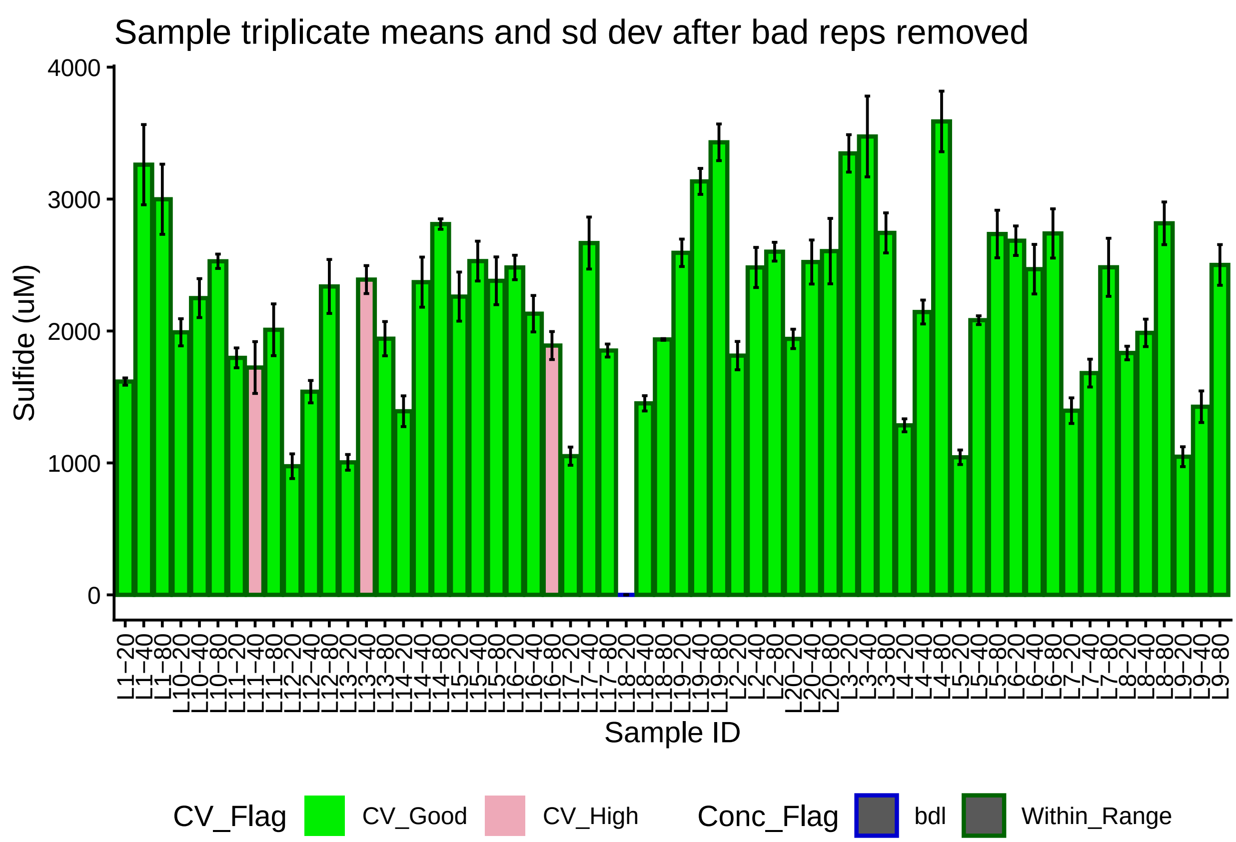
Plate	CHK_CV_Flags
Plate2	Std4Chk_CV_High

##Calculate Sulfide Concentrations & Add Concentration Flags

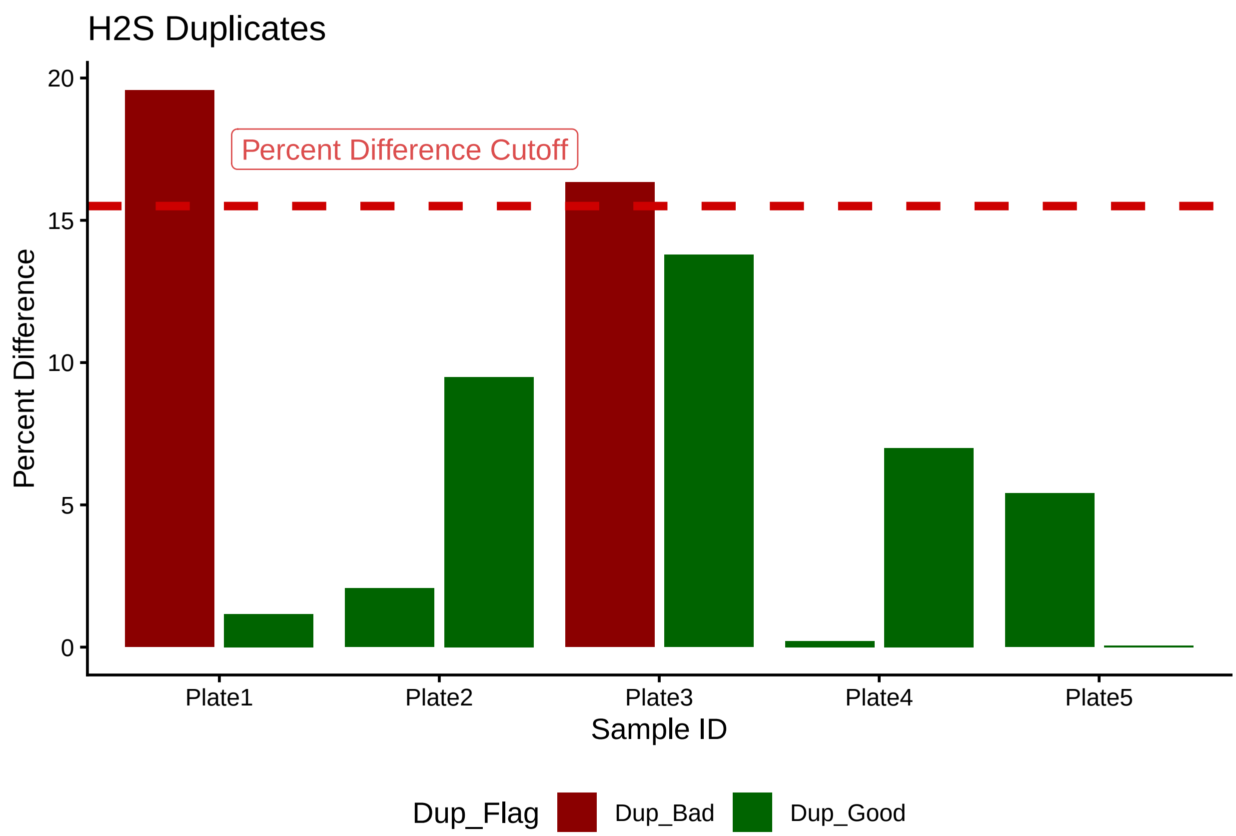
##Remove High CV Samples and Average

Plot Samples





Duplicate QAQC

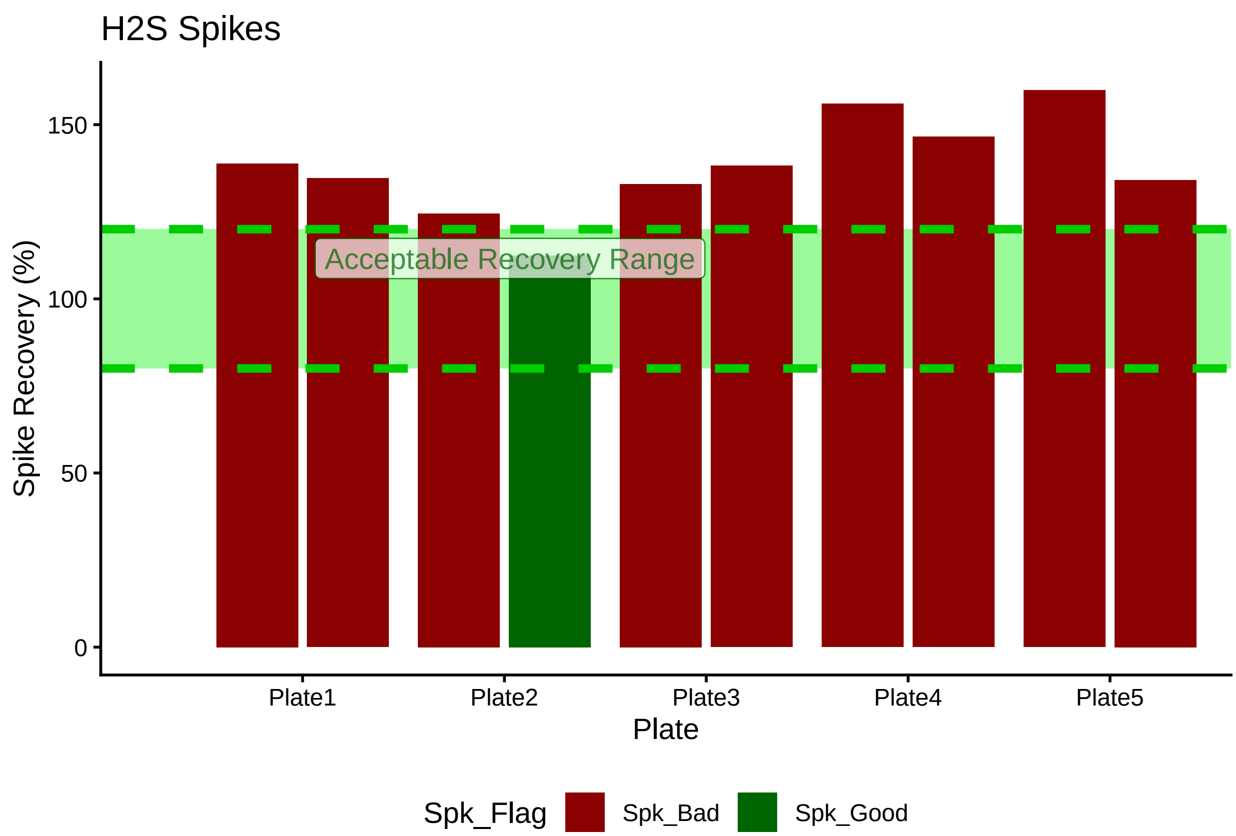


Display Dup Flags

```
## [1] ">60% Dups Pass"
```

Plate	Dup_Flag
Plate1	Dup_Bad
Plate2	Dup_Good
Plate3	Dup_Bad
Plate4	Dup_Good
Plate5	Dup_Good

Spike QAQC



Display Spike Flags

```
## [1] "Spikes Paired"
```

Plate	Spike_Flag
Plate1	Spk_Bad
Plate2	Spk_Bad
Plate3	Spk_Bad
Plate4	Spk_Bad
Plate5	Spk_Bad

```
## [1] "<60% of Spikes Pass, Rerun"
```

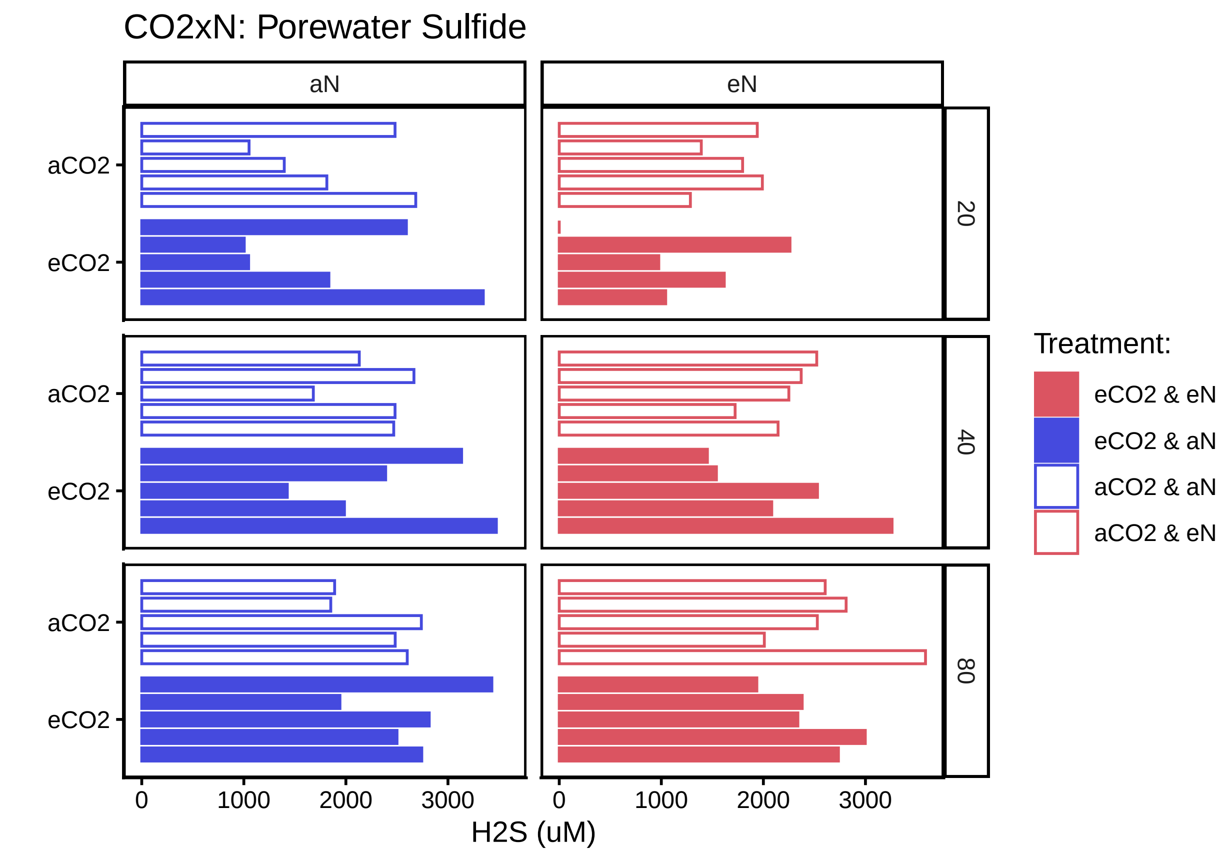
Merge samples with Metadata & check all samples are present

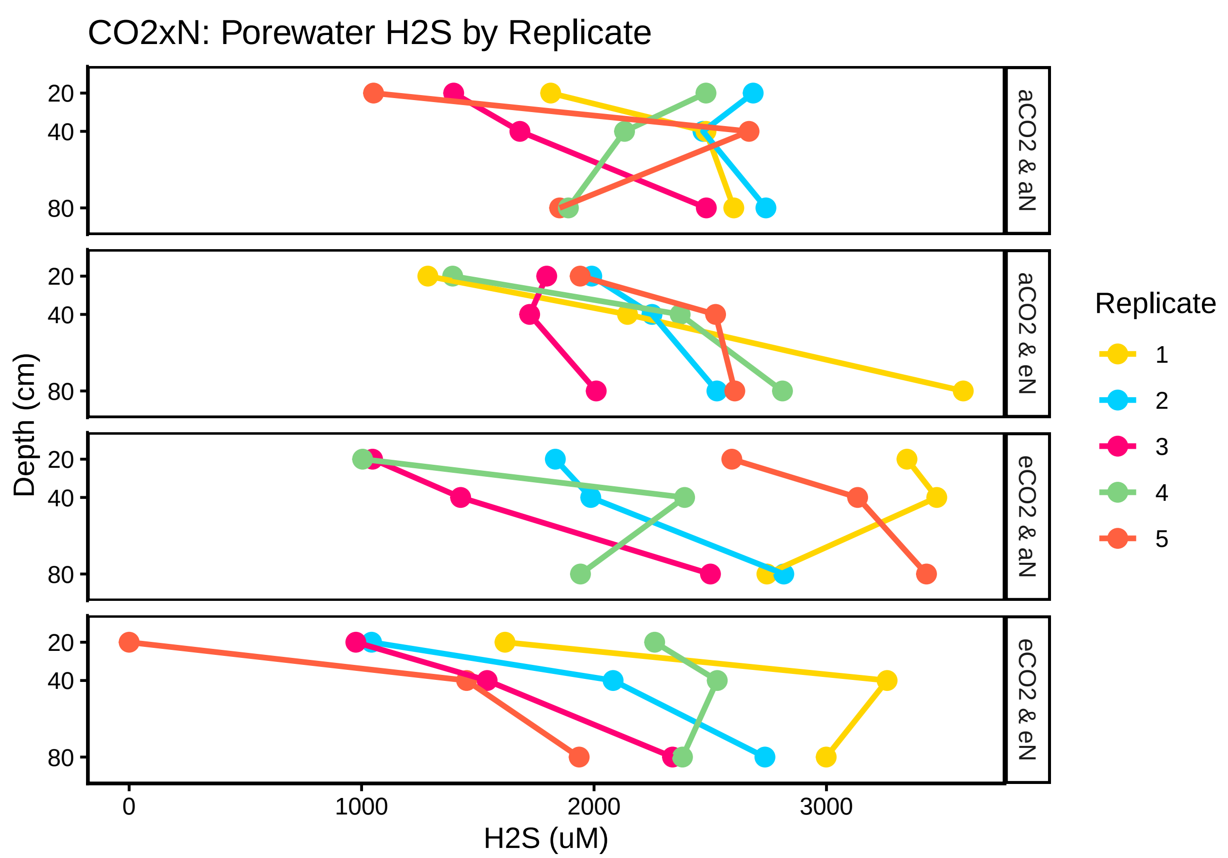
```
## All sample IDs are present in data
```

##Add Flags

##Organize Data

Visualize Data





##Export Data

END